

Study on Accuracy of Trilateration Method for Indoor Positioning with BLE Beacons

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Abstract— Nowadays, indoor positioning services of mobile phone with Bluetooth low energy (BLE) beacons are increasingly used. However, due to the indoor environment has dense multipath fading, the positioning system has low accuracy. In this paper, the accuracy study of trilateration method for indoor positioning with BLE beacons is presented. The indoor environment room with 9.0 m width, 10.2 m length and 3 BLE beacons is studied. The received signal strength (RSS) of each BLE beacon is measured by using mobile phone. The distance between each BLE beacon and mobile phone is evaluated using channel model. After that, the trilateration method is used to approximate the position of mobile phone. The accuracies, which are obtained from one, average five and average ten measurements, are studied. The distance errors of all positions and the cumulative distribution function (CDF) of the distance error are calculated and shown. The results show that the average multiple measurements tend to decrease the distance error.

Keywords—Indoor positioning, trilateration method, BLE beacon.

I. INTRODUCTION

Nowadays, mobile indoor location positioning services have started to use Bluetooth low energy (BLE) beacons, because it's cheap, easy to install and can develop through a program in the mobile phone [1]. There is the research about indoor positioning with BLE beacons using different methods. For example, positioning that uses path loss models [2], weighted centroid approach [3] and fingerprinting techniques [4]-[5]. However, the indoor positioning with BLE beacons has high error because the indoor environment often has dense multipath fading. Therefore, the study of accuracy is essential to improve positioning technique for decreasing error.

In this paper, the accuracy study of the trilateration method for indoor positioning with BLE beacons is presented. The indoor environment is the 9.0 m width and 10.2 m length room with 3 BLE beacons. The Received signal strength (RSS) of each BLE beacon is measured by using mobile phone. The distance between each BLE beacon and mobile phone is approximated by using the channel model [6]. Then, the trilateration method is used to find mobile phone location. The accuracies, which are obtained from one, average five and average ten measurements, are studied. The distance errors of all position and the cumulative distribution function (CDF) of distance error are calculated and shown. The results show that the average multiple measurements can reduce the distance error.

The rest of paper consists as follow. Section II is the trilateration method. After that, the measurement model is described in Section III. Next, the study of accuracy is shown in Section IV. Finally, Section V is conclusion.

II. TRILATERATION METHOD

The trilateration method is a popular method for positioning. This method requires at least 3 transmitters. In this paper, the BLE beacons are use as transmitters. The distance between each BLE beacons and mobile phone is estimated. After that, these distances are defined as the radius of the circles and the positions of BLE beacons are defined as the center of circles. The circumferences of these circles are all possible positions of mobile phone. In theory, these circles intersect at one point, which is the mobile phone position.

Positioning using the trilateration method is shown in Fig. 1, where the positions of BLE beacons 1, 2, and 3 are (x_1, y_1) , (x_2, y_2) and (x_3, y_3) , respectively. The parameters d_1 , d_2 , and d_3 are the distances between BLE beacons 1, 2 and 3, and mobile phone. The position of mobile phone is (x_e, y_e) .

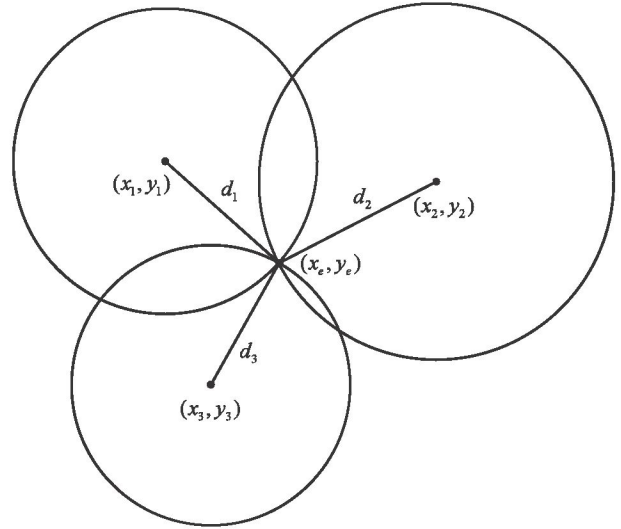


Fig. 1. Positioning using the trilateration method.

In practice, because of heavy multipath fading in the indoor environment, the estimation of distances d_1 , d_2 , and d_3 causes error. In this paper, these distances are estimated by using the channel model proposed in [6]. The obtained distances with error cause the circles to not intersect at one point. In addition, the position of mobile phone cannot be calculated directly by solving the equations of circles. Therefore, the least square error method is applied to estimate the position of mobile phone.

First, the 3 circular equations, which have their centers of (x_1, y_1) , (x_2, y_2) and (x_3, y_3) positions and their radius of d_1 , d_2 , and d_3 distances, are considered. These circular equations can be expressed as

$$(x_e - x_1)^2 + (y_e - y_1)^2 = d_1^2, \quad (1)$$

$$(x_e - x_2)^2 + (y_e - y_2)^2 = d_2^2, \quad (2)$$

$$(x_e - x_3)^2 + (y_e - y_3)^2 = d_3^2. \quad (3)$$

Next, the squares in all above equations can be expanded and rearranged as

$$x_e^2 - 2x_1x_e + y_e^2 - 2y_1y_e = d_1^2 - x_1^2 - y_1^2, \quad (4)$$

$$x_e^2 - 2x_2x_e + y_e^2 - 2y_2y_e = d_2^2 - x_2^2 - y_2^2, \quad (5)$$

$$x_e^2 - 2x_3x_e + y_e^2 - 2y_3y_e = d_3^2 - x_3^2 - y_3^2. \quad (6)$$

After that, taking two of three above equations and subtracting them together get all possible linear equations as follow

$$(4)-(5) \quad A_1x_e + B_1y_e = C_1, \quad (7)$$

$$(4)-(6) \quad A_2x_e + B_2y_e = C_2, \quad (8)$$

$$(5)-(6) \quad A_3x_e + B_3y_e = C_3, \quad (9)$$

where A_1 , A_2 , A_3 , B_1 , B_2 , B_3 , C_1 , C_2 and C_3 are defined as

$$A_1 = 2(x_2 - x_1),$$

$$A_2 = 2(x_3 - x_1),$$

$$A_3 = 2(x_3 - x_2),$$

$$B_1 = 2(y_2 - y_1),$$

$$B_2 = 2(y_3 - y_1),$$

$$B_3 = 2(y_3 - y_2),$$

$$C_1 = d_1^2 - d_2^2 + x_2^2 - x_1^2 + y_2^2 - y_1^2,$$

$$C_2 = d_1^2 - d_3^2 + x_3^2 - x_1^2 + y_3^2 - y_1^2,$$

$$C_3 = d_2^2 - d_3^2 + x_3^2 - x_2^2 + y_3^2 - y_2^2.$$

In practice, the distances d_1 , d_2 and d_3 , which is obtained from channel model, have error. Consequently, solving each pair of linear equations Eqs. (7)-(9) do not

obtain the identical answer. Therefore, the least square error method is used to approximate the mobile phone position (x_e, y_e) . The square error of Eqs. (7)-(9) can be written as

$$E_1^2 = (C_1 - A_1x_e - B_1y_e)^2 \quad (10)$$

$$E_2^2 = (C_2 - A_2x_e - B_2y_e)^2 \quad (11)$$

$$E_3^2 = (C_3 - A_3x_e - B_3y_e)^2 \quad (12)$$

The total square error can be found from

$$\begin{aligned} E_1^2 + E_2^2 + E_3^2 \\ = (C_1 - A_1x_e - B_1y_e)^2 + (C_2 - A_2x_e - B_2y_e)^2 \\ + (C_3 - A_3x_e - B_3y_e)^2. \end{aligned} \quad (13)$$

The mobile phone position (x_e, y_e) is estimated from x_e and y_e that make Eq. (13) with least square error. It obtains by taking the partial derivative of Eq. (13) with respect to x_e and y_e and then equal to 0, which are

$$\begin{aligned} \frac{\partial}{\partial x_e} [(C_1 - A_1x_e - B_1y_e)^2 + (C_2 - A_2x_e - B_2y_e)^2 \\ + (C_3 - A_3x_e - B_3y_e)^2] = 0, \end{aligned} \quad (14)$$

$$\begin{aligned} \frac{\partial}{\partial y_e} [(C_1 - A_1x_e - B_1y_e)^2 + (C_2 - A_2x_e - B_2y_e)^2 \\ + (C_3 - A_3x_e - B_3y_e)^2] = 0. \end{aligned} \quad (15)$$

The results of partial derivative of Eqs. (14) and (15) are

$$\begin{aligned} (C_1 - A_1x_e - B_1y_e)A_1 + (C_2 - A_2x_e - B_2y_e)A_2 \\ + (C_3 - A_3x_e - B_3y_e)A_3 = 0, \end{aligned} \quad (16)$$

$$\begin{aligned} (C_1 - A_1x_e - B_1y_e)B_1 + (C_2 - A_2x_e - B_2y_e)B_2 \\ + (C_3 - A_3x_e - B_3y_e)B_3 = 0. \end{aligned} \quad (17)$$

Equations (16) and (17) can be rearranged as

$$\begin{aligned} (A_1^2 + A_2^2 + A_3^2)x_e + (A_1B_1 + A_2B_2 + A_3B_3)y_e \\ = A_1C_1 + A_2C_2 + A_3C_3, \end{aligned} \quad (18)$$

$$\begin{aligned} (A_1B_1 + A_2B_2 + A_3B_3)x_e + (B_1^2 + B_2^2 + B_3^2)y_e \\ = B_1C_1 + B_2C_2 + B_3C_3. \end{aligned} \quad (19)$$

The mobile phone position (x_e, y_e) can be evaluated from

$$\begin{aligned} \begin{bmatrix} x_e \\ y_e \end{bmatrix} = \begin{bmatrix} A_1^2 + A_2^2 + A_3^2 & A_1B_1 + A_2B_2 + A_3B_3 \\ A_1B_1 + A_2B_2 + A_3B_3 & B_1^2 + B_2^2 + B_3^2 \end{bmatrix}^{-1} \\ \times \begin{bmatrix} A_1C_1 + A_2C_2 + A_3C_3 \\ B_1C_1 + B_2C_2 + B_3C_3 \end{bmatrix} \end{aligned} \quad (20)$$

The distance error is considered for studying the accuracy of indoor positioning. The distance error is calculated from the absolute difference between correct position (x_c, y_c) and estimated position (x_e, y_e) . The distance error along x and y axes can be written as

$$E_x = |x_c - x_e|, \quad (21)$$

$$E_y = |y_c - y_e|. \quad (22)$$

Distance error E_d is defined as the magnitude of vector sum of distance error along x and y axes, which is

$$E_d = \sqrt{E_x^2 + E_y^2}. \quad (23)$$

This distance error obtained from one, average five and average ten measurements is studied in the term of CDF.

III. MEASUREMENT MODEL

The indoor environment is computer laboratory room, 3rd floor, Department of Telecommunications Engineering, Faculty of Engineering, King Mongkut's Institute of Technology Ladkrabang. This room has 9.0 m width and 10.2 m length. Therefore, the total area of this room is 91.8 m². The measurement positions are divided into 15 points along width (x axis) and 17 points along length (y axis) with 0.6 m space between each point. The total measurements positions are 163 points. The 3 BLE beacons are attached at right, left and back sides of this room. All BLE beacons are at 1.8 m height from the floor. The RSS of each BLE beacon along each position is measured by using mobile phone.

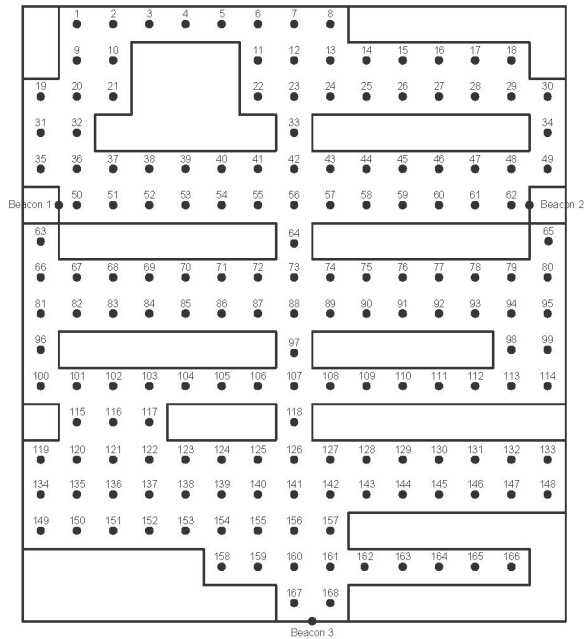


Fig. 2. The layout of measurement room.

The measurements are divided into 3 cases which are one, average five and average ten measurements. The obtained RSS is used to estimate distance between mobile phone and BLE beacons using the channel model presented in [6] and estimate the position of mobile phone using the trilateration method. The layout of this room that used for indoor positioning is shown in the Fig. 2.

IV. ACCURACY STUDY

Channel model presented in [6] is shown in the Fig. 3. This channel model with path loss exponent $n=1.27$ and The RSS at 1 m distance is $P_r(1\text{m}) = -69.72$ dBm is used to estimate the distance between mobile phone and BLE beacons d by using

$$d = 10^{[P_r - P_r(1\text{m})]/(-10n)}, \quad (24)$$

where P_r is the RSS of BLE beacons.

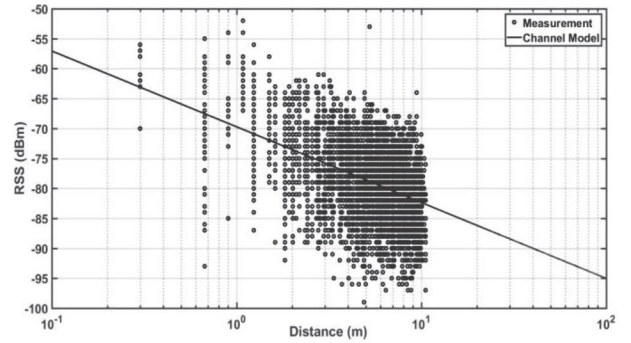


Fig. 3. Channel model [6]

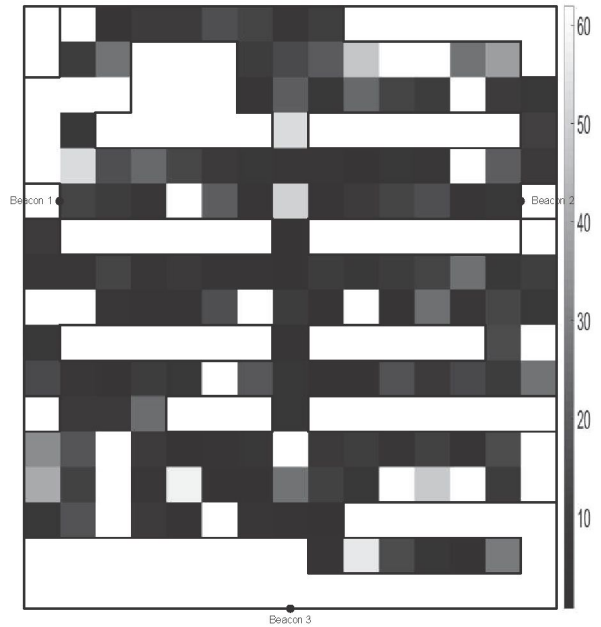


Fig. 4. The distance error obtained from one measurement.

Figures 4 to 6 show the distance error at each position obtained from one, average five, and average ten measurements, respectively. The dark position indicates low error. On the other hand, the bright position indicates high error. The color bar, which is at right side, is used for comparing RSS with unit of dBm. From the results, the distance error obtained from one measurement is maximum. The distance error obtained from average five measurements is less. Furthermore, the distance error obtained from average ten measurements is minimum.

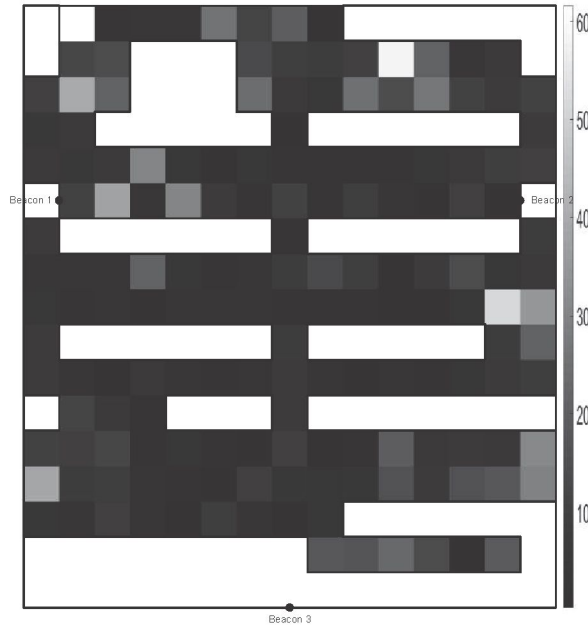


Fig. 5. The distance error obtained from average five measurements.

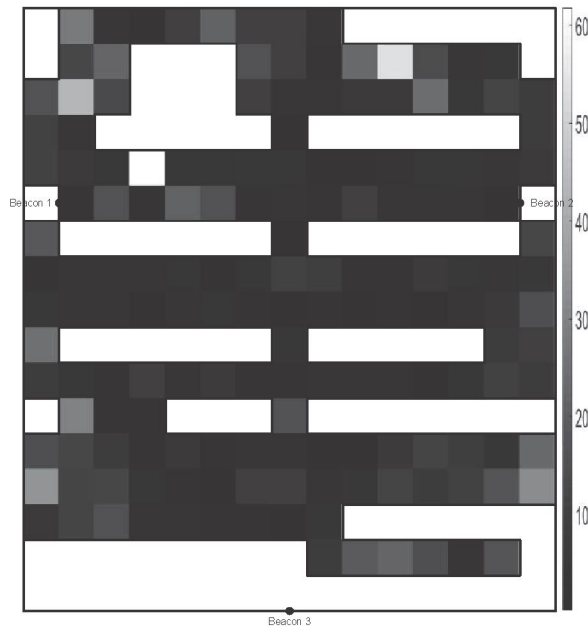


Fig. 6. The distance error obtained from average ten measurements.

Figure 7 shows the CDF of distance error obtained from one, average five and average ten measurements. The means of distance error are 39.06 m, 8.96 m and 7.97 m respectively. The medians of distance error (CDF = 0.5) are 8.04, 5.54, 4.23 m, respectively.

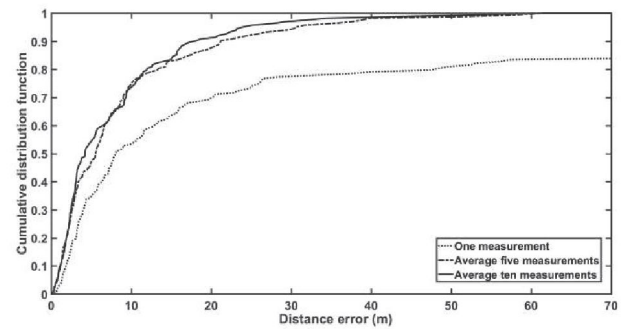


Fig. 7. CDF of distance error obtained from one, average five and average ten measurements.

V. CONCLUSION

In this paper, the study of the accuracy of the trilateration method for indoor positioning with BLE beacons is presented. The accuracy obtained from one, average five and average ten measurements are studied. From the results, when the number of measurements is increased, the distance error tends to decrease. The multiple measurements of RSS can be used to improve the indoor position accuracy because the average RSS decreases the effect of multipath fading that occurred in the indoor environment.

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